www.PapaCambridge.com

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2010 question paper

for the guidance of teachers

4024 MATHEMATICS (SYLLABUS D)

4024/12 Paper 1, maximum raw mark 80

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- CIE will not enter into discussions or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the October/November 2010 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.

www.PapaCambridge.com

Abbreviations

cao correct answer only
cso correct solution only
dep dependent
ft follow through after error
isw ignore subsequent working
oe or equivalent
SC Special Case
www without wrong working
art anything rounding to
soi seen or implied

1	**(a)**	$\frac{17}{21}$ oe	1	
	(b)	$\frac{5}{12}$ cao	1	
2	**(a)**	70	1	
	(b)	4.05	1	
3	**(a)**	7.06×10^{-5} cao	1	
	(b)	150	1	
4	**(a)**	7	1	
	(b)	6	1	
5	**(a)**	1.65	1	
	(b)	2 : 25	1	
6	**(a)**	$(2t - 3)(2t + 3)$	1	
	(b)	$(3x - 1)(x + 2)$	1	
7	18		2	or **B1** for "k" = 2, or for $\frac{y}{50} = \frac{3^2}{5^2}$ oe
8	$(\pm)\sqrt{\frac{y-3}{2}}$ oe e.g. $(\pm)\left(\frac{y-3}{2}\right)^{\frac{1}{2}}$ (sq. root symbol must extend below the fraction line)		2	or **C1** for $\frac{\sqrt{y-3}}{2}$ or for $\sqrt{\frac{y+3}{2}}$ or for $\sqrt{\frac{3-y}{2}}$ or for $\sqrt{y-3/2}$ oe for all
9	**(a)**	(±) 5 cao	1	
	(b)	**(i)** 6	1	
		(ii) (1.5, 0)	1	

www.PapaCambridge.com

10	**(a)**	$\frac{4}{5}$, or 0.8, only	1	
	(b)	$25x^6$ cao	1	
	(c)	$\frac{4}{n^8}$	1	
11	**(a)**	8	1	
	(b)	{5, 6, 7, 8, 9}	1	
	(c)	$\frac{3}{10}$ or 0.3	1	
12	**(a)**	$3\frac{1}{2}$, or $\frac{7}{2}$, or 3.5, only	1	
	(b)	$12 - 2x$ or **any** equivalent	2	or **C1** for $12 - 2$ "y" or **any** equivalent or **C1** for $6 - 2x$, or for any incorrect linear combination of 12 and $2x$ (but not 2"y")
13	**(a)**	Irrational	1	
	(b)	$(AB^2 =)\ AC^2 - 5^2$ or $(AB =)\ \sqrt{AC^2 - 5^2}$ or $AC^2 = AB^2 + 5^2$. AC must be "their" $\sqrt{89}$	M1	
		$(\pm)\ 8$	A1	
14	$x = 9,\ y = 6$ both		3	or **C2** for one answer correct; or **C1** for a pair of values that fits either equation, provided that this pair has been obtained by the method of substitution, equal coeffs., or matrices/determinants and **not** by trial and error.
15	**(a)**	16 (.0)(0)	1	
	(b)	75 (.0)(0) www	2	or **M1** for $\frac{60}{0.8}$ oe, e.g. $\frac{3k \times 100}{4k}$
16	**(a)**	$\begin{pmatrix} -1 & -2 \\ 0 & -2 \end{pmatrix}$	1	
	(b)	$\begin{pmatrix} 0 & -1 \\ -\frac{1}{3} & -\frac{2}{3} \end{pmatrix}$ oe e.g. $-\frac{1}{3}\begin{pmatrix} 0 & 3 \\ 1 & 2 \end{pmatrix}$	2	or **B1** for det **A** = –3 or for $k\begin{pmatrix} 0 & 3 \\ 1 & 2 \end{pmatrix}$ or for $-\frac{1}{3}\begin{pmatrix} .. & .. \\ .. & .. \end{pmatrix}$
17	**(a)**	$\begin{pmatrix} 490 \\ 520 \end{pmatrix}$	2	or **C1** for a 2×1 matrix with one element correct; or for (490 520)
	(b)	The cost, (in cents), of each bunch. oe	1	Indep. of **(a)**
18	**(a)**	14.7(0) cao	1	
	(b)	30	2	or **B1** for 170 seen in working or in Ans. Space

www.PapaCambridge.com

19	**(a)**	$p = 1,\ q = 0$	1	
	(b)	**(i)** $\frac{5}{7}$	1	
		(ii) $\frac{2}{7}$ or ft 1 – their **(i)** or ft $(\frac{1}{7}+\frac{1}{7} \times$ their $p)$	1ft	ft depends on 0 < Ans. < 1
20	**(a)**	$3x > 7$ oe $4x + 4y < 35$ oe	1 1	or **C1** for $3x$ ... 7 **and** $4x + 4y$ ... 35 (oe) with incorrect inequalities for ... .
	(b)	(5, 3)	1	
21	**(a)**	53.35°	1	
	(b)	65.15°	2	or **C1** for 64.65; or 65.1; or 64.05
22	**(a)**	**(i)** 16 000 cao **(ii)** 0.0030 cao	1 1	
	(b)	50 cao	2	Give 0 for multiplication using either original number. or **C1** for figs. 5, or 6; or 45; or 48
23	**(a)**	123°	1	
	(b)	57°	1	
	(c)	33°	1	
	(d)	66°	1	
24	**(a)**	$3\mathbf{p} + \mathbf{q}$ oe	1	
	(b)	**(i)** Trapezium **(ii)** $\mathbf{p} + k\mathbf{q}$ oe **(iii)** $\frac{1}{3}$	1 1 1	
25	**(a)**	30	2	or **B1** for $10u$ or ½ × 20 × u clearly seen
	(b)	90	2	or **C1** for 30 (if as the further time from 60) or **M1** for $100 - \frac{1}{4} \times 40$, or for $60 + \frac{3}{4} \times 40$
26	**(a)**	$-\frac{4}{5}$, or –0.8, only	1	
	(b)	16	3	or **M1** for $\frac{AC}{\sin b} = \frac{10}{\sin a}$ soi and **M1** for $AC = \frac{10 \times \frac{24}{25}}{\frac{3}{5}}$ oe

www.PapaCambridge.com

27	**(a)**	3	1	
	(b)	80	1	
	(c)	$7\frac{1}{2}$ oe	3	or **B1** for $\frac{A}{360} \times 2 \times \pi \times r$ with $A = 40$ or 120, **and** $\pi = \pi$ or 3 or 3.14 or $\frac{22}{7}$ etc and **M1** for adding the appropriate radii ($6r$ or $2r$) to their arc(s) **and** equating to the appropriate wire length (60 or 20) ($8r = 60$, oe (e.g. $\frac{8}{3}r = 20$), gets B1 and M1)